

IA-Based IDS

What is the difference between how a **classical machine** and a **human** perform a task?

- While machines excel at **speed, accuracy, and repetition,**
- humans have strengths in **reasoning, emotions, creativity, and adaptability (making Decisions).**

What is Artificial Intelligence?

The simulation of human intelligence in machines, enabling them to perform tasks like **learning**, **reasoning**, and **problem-solving**.

Examples:

- ✓ ChatGPT
- ✓ Siri
- ✓ facial recognition

AI vs ML vs DL

- AI: Broad field
 - Machine Learning (ML): Subfield of AI
 - Deep Learning (DL): Subfield of ML

Real-world Applications of AI

- Healthcare
 - Cancer detection
- Industry
 - Autonomous Cars



Source: *Saturday Evening Post*, 1956



Tesla

Waymo



Real-world Applications of AI

- Social media
 - Recommendation systems
- Cybersecurity
 - malware detection, phishing prevention, anomaly detection.

What is Machine Learning?

A subset of AI where **systems learn** patterns from **data without explicit programming**.

Core Idea:

- "ML algorithms **improve automatically through experience**."

It is like « Teaching a child to recognize dogs vs cats by showing examples or Fruit classification based on colors and weight to predict fruit type »

"Machine Learning learns from data – but how?"

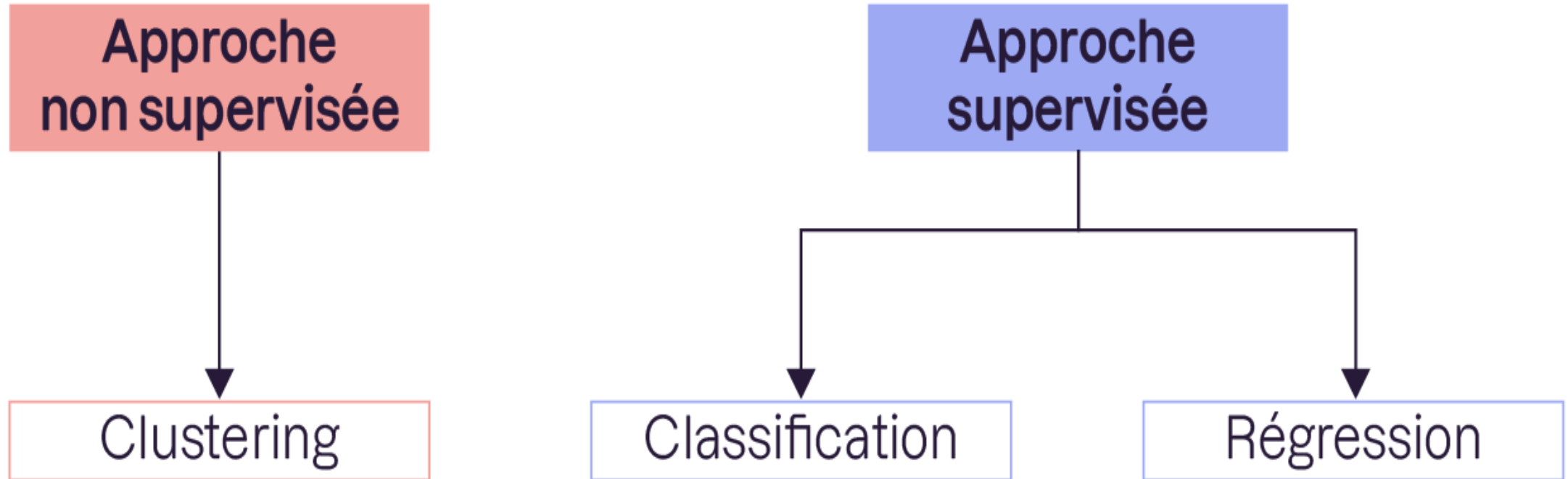
▶ Supervised Learning: The model learns patterns from examples where the "correct answer" is provided.

(labeled data → predictions)

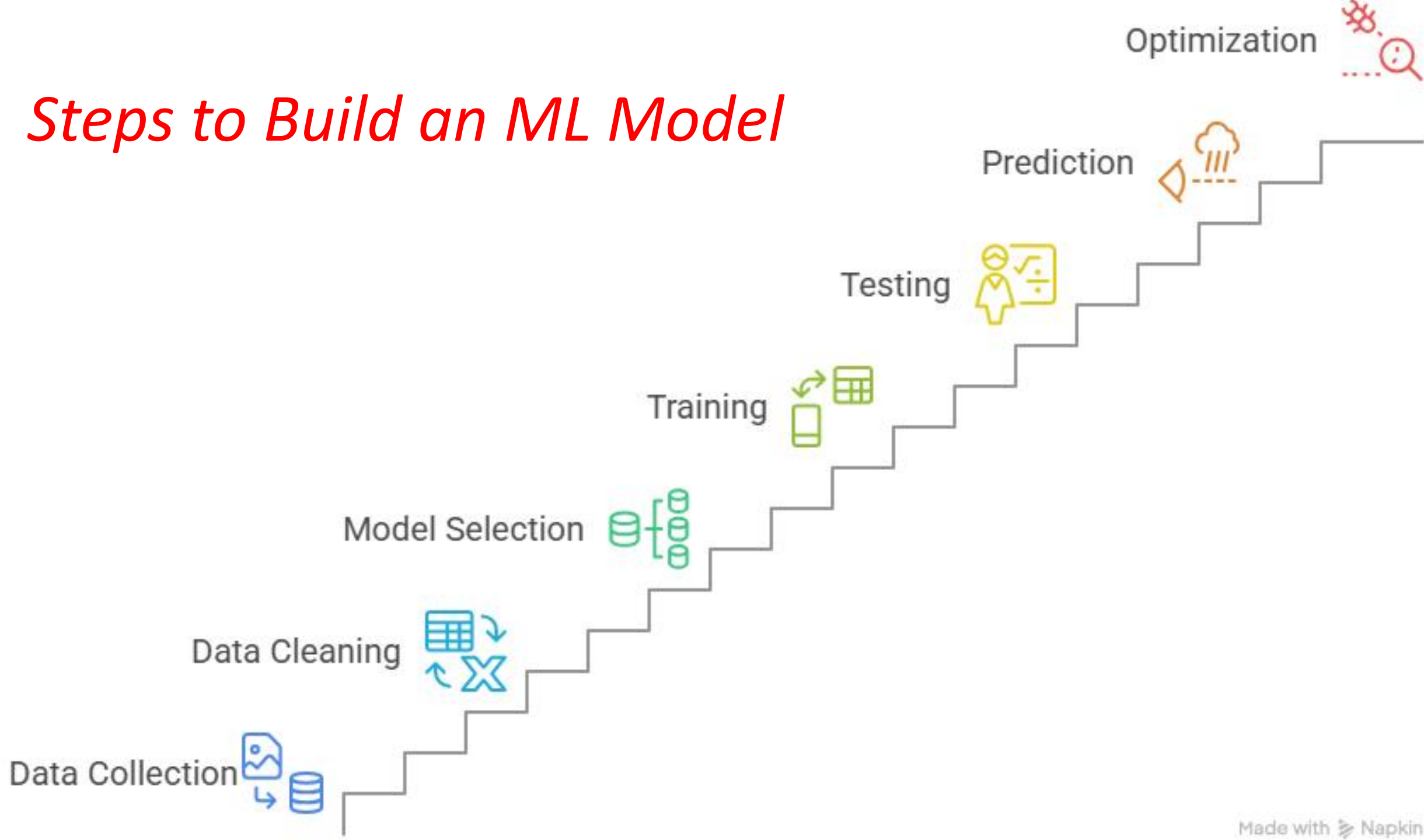
▶ Unsupervised Learning: The model explores data structure on its own to Find the hidden patterns.

(no labels → discovery)

"Machine Learning learns from data – but how?"



Steps to Build an ML Model



Steps to Build an ML Model

1. Data collection : Download (csv) or Generate ?

Steps to Build an ML Model

2. Data cleaning:

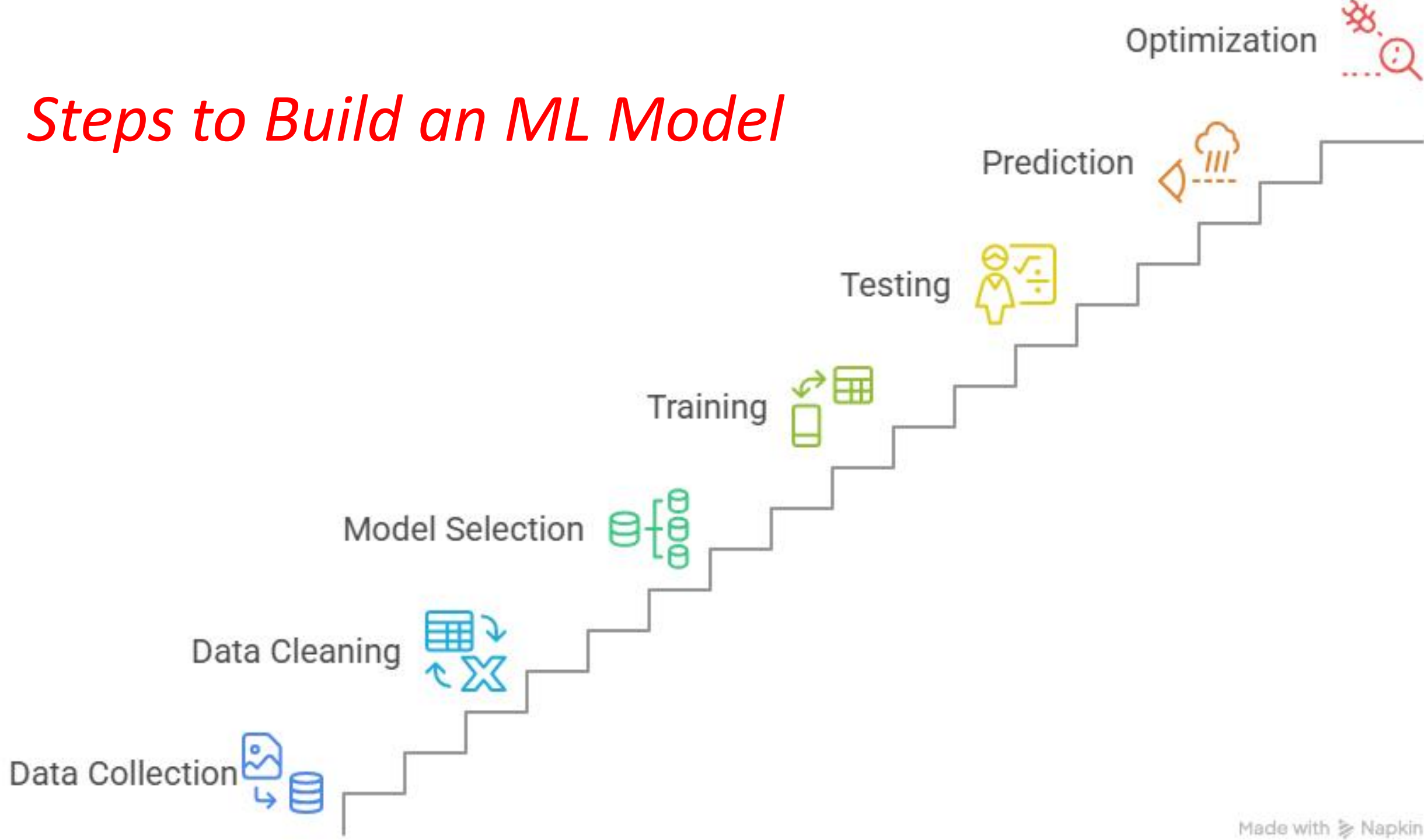
The process of detecting and correcting (or removing) errors, inconsistencies, and noise in datasets to improve quality for ML models.

Issue	Example	Solution
Missing values	Blank cells in a spreadsheet	Fill/replace or drop rows
Duplicates	Same customer entered twice	Remove redundant entries
Outliers	Age = "150" in a patient dataset	Cap values or investigate
Inconsistent formatting	Dates as "Jan 2023" vs. "01/23"	Standardize formats, Hot Encoding

Reminder

1. IA?
2. ML?
3. Supervised vs Unsupervised ?
4. Output ?
5. DL ?

Steps to Build an ML Model



Steps to Build an ML Model

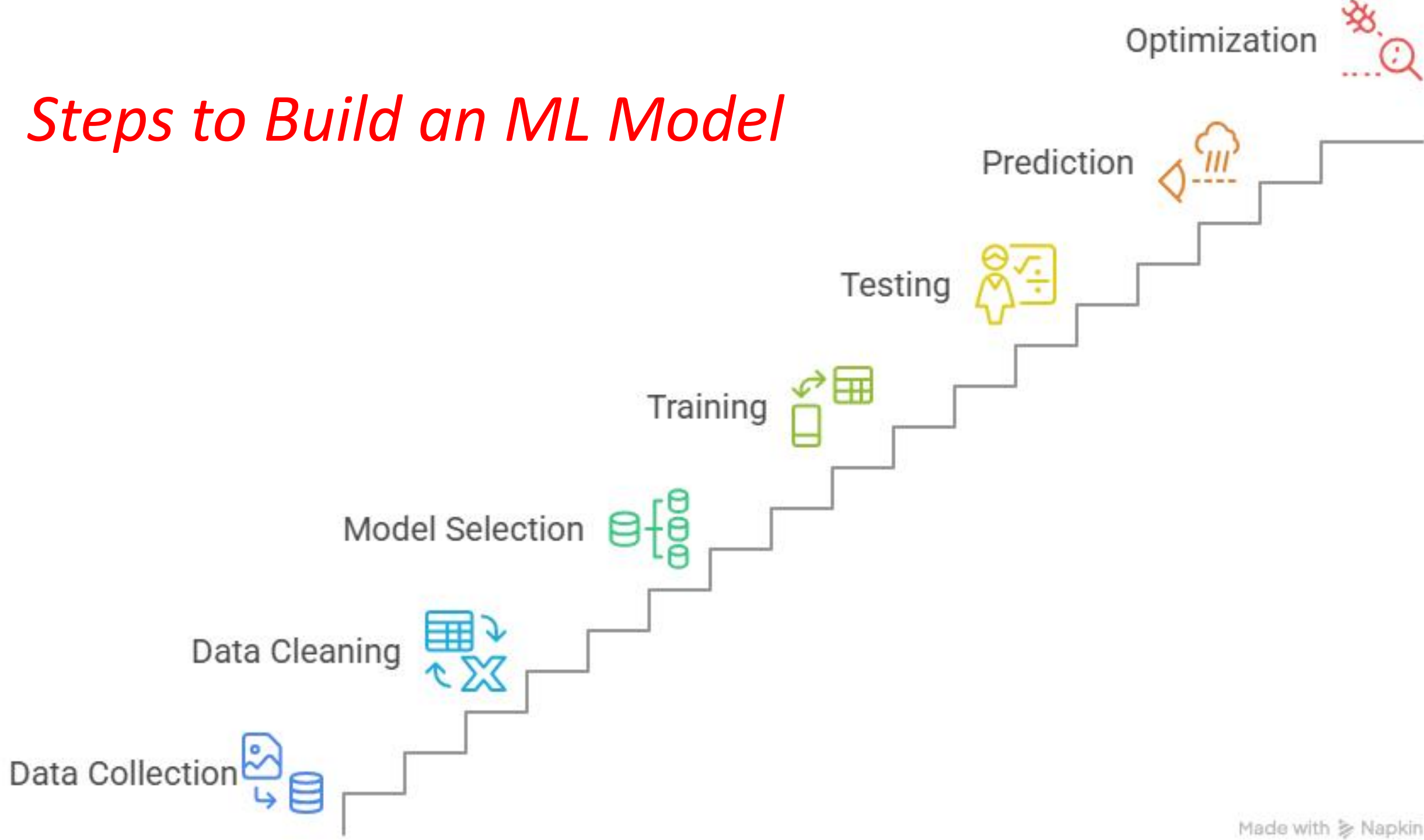
3. Model selection:

The process of choosing the best machine learning algorithm(s) for a specific problem, based on data type, task, and performance goals.

Key Considerations for Model Selection

1. Problem Type: *Is it prediction, classification, or clustering?*
2. Data Size & Complexity ?
3. Speed, Lightweight vs. Accuracy

Steps to Build an ML Model



Steps to Build an ML Model

4. Training & Test: *Why Split Data?*

To evaluate if your model generalizes well to new, unseen data (avoid overfitting).

1. Training Set (80% of Data):

Purpose: Teach the model patterns.

Process: Algorithm learns from features (X) and labels (y).

Example:

- **Input (X):** Humidity, Temp, Wind
- **Output (y):** Play Tennis (labeled data).

2. Test Set (20% of Data):

Purpose: Simulate real-world performance.

Rule: Never let the model "see" this during training!

Metric Examples:

Classification: Accuracy, Precision, Recall.

Accuracy:

The **ratio of correct predictions** (both positive and negative) **to the total** number of predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

Of all the samples the model predicted as positive (attack), how many were actually positive?

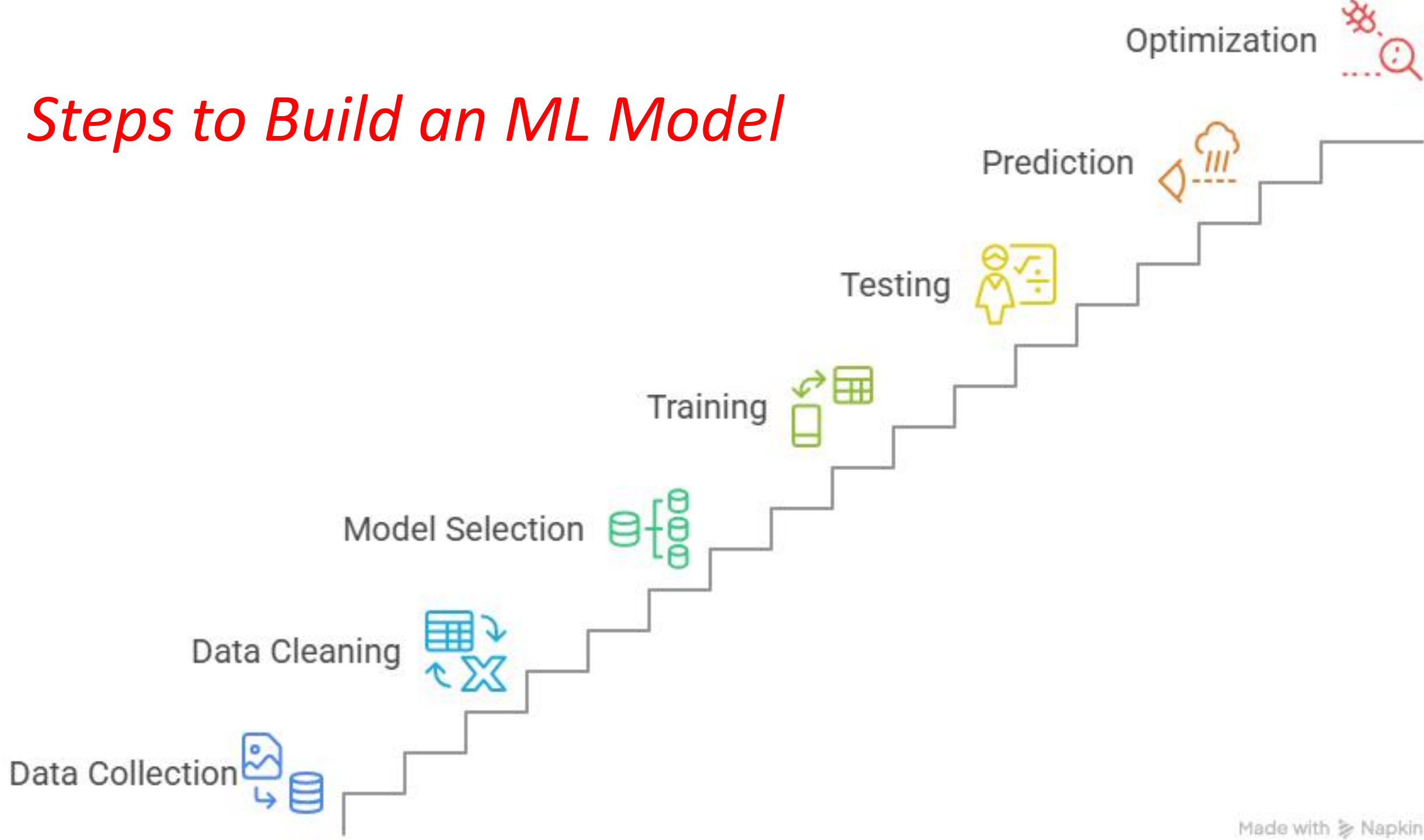
$$Precision = \frac{TP}{TP + FP}$$

Recall:

Of all the actual positives, how many did the model correctly detect?

$$Recall = \frac{TP}{TP + FN}$$

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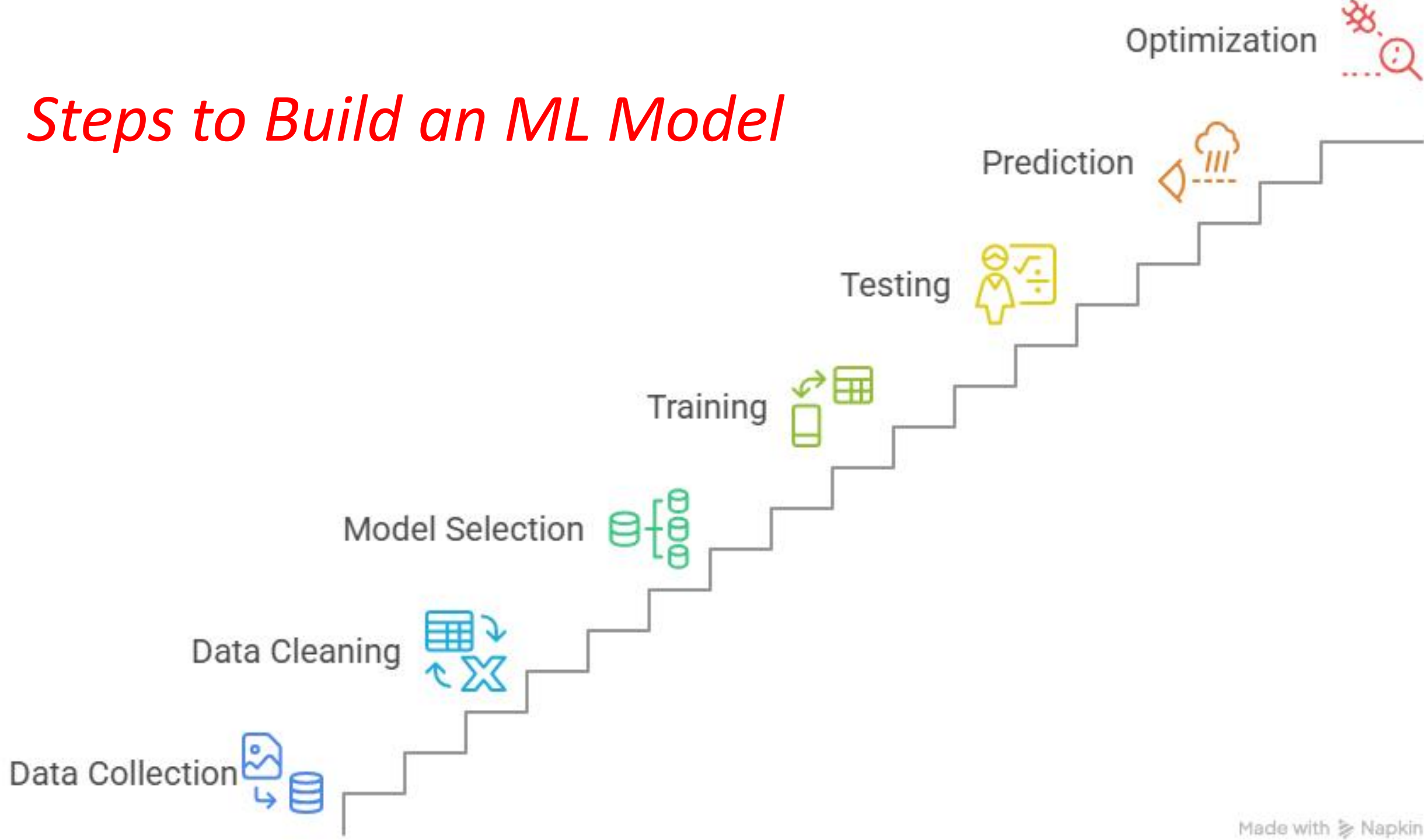


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6. Prediction ?

The moment of truth where models meet real-world data

Steps to Build an ML Model



Steps to Build an ML Model

7. Optimisation ?

The process of:

1. Maximizing accuracy/reliability of predictions
2. Minimizing errors and computational costs

Key Areas of Optimization:

1. Hyperparameter Tuning : Adjusting model settings before training
2. Model Architecture
3. Feature Optimization

Steps to Build an ML Model

1. Data collection
2. Data cleaning
3. Model selection
4. Training
5. Test
6. Prediction
7. Optimisation

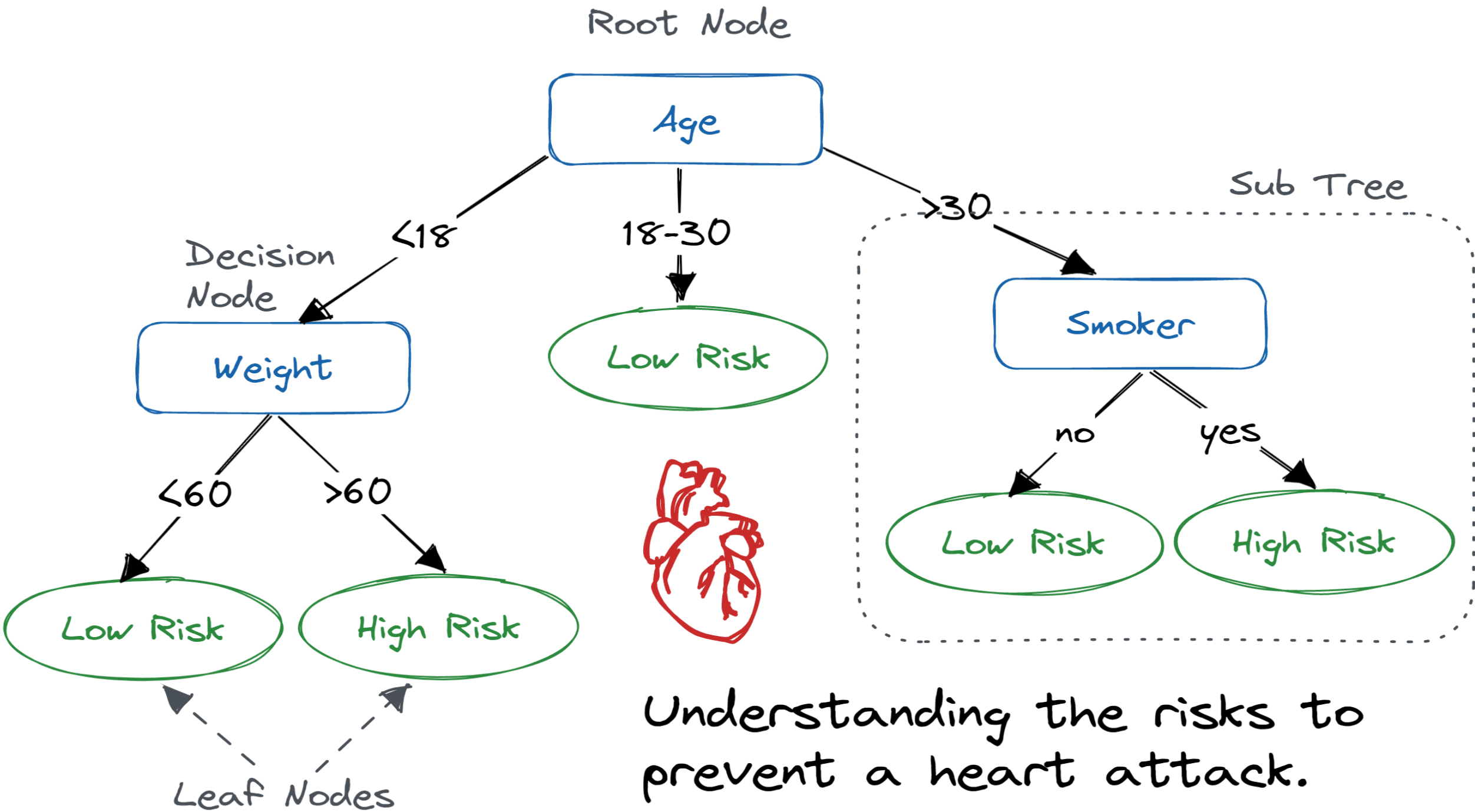
Use Case: Decision Tree Machine Learning Algorithm

How Does It Work?

What is a Decision Tree?

A decision tree is a Tree-based structure model used for classification and regression tasks, where data is split into branches based on decision rules to reach a predicted outcome.

Each node = Question/Condition.



How does it work?

Entropy

Entropy is a measure of impurity or disorder in a dataset. In decision trees, it quantifies the uncertainty in predicting the class of a randomly chosen instance.

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \cdot \log_2(p_i)$$

- S : the dataset
- c : the number of classes (in our example, 2 classes: Yes and No)
- p_i : the proportion of elements belonging to class i in the dataset

l'entropie globale (Entropy(S)) :

$$\text{Entropy}(S) = 0.940$$

📌 Cas 1 : Humidity = High

Exemples : 7

- 3 "Yes"
- 4 "No"

$$p(\text{Yes}) = \frac{3}{7}, \quad p(\text{No}) = \frac{4}{7}$$

$$\begin{aligned}\text{Entropy}(\text{High}) &= - \left(\frac{3}{7} \log_2 \frac{3}{7} + \frac{4}{7} \log_2 \frac{4}{7} \right) \\ &= -(0.429 \cdot \log_2 0.429 + 0.571 \cdot \log_2 0.571) \approx 0.985\end{aligned}$$

📌 Cas 2 : Humidity = Normal

Exemples : 7

- 6 "Yes"
- 1 "No"

$$p(\text{Yes}) = \frac{6}{7}, \quad p(\text{No}) = \frac{1}{7}$$

$$\text{Entropy}(\text{Normal}) = - \left(\frac{6}{7} \log_2 \frac{6}{7} + \frac{1}{7} \log_2 \frac{1}{7} \right)$$

$$= -(0.857 \cdot \log_2 0.857 + 0.143 \cdot \log_2 0.143) \approx 0.592$$

Information Gain

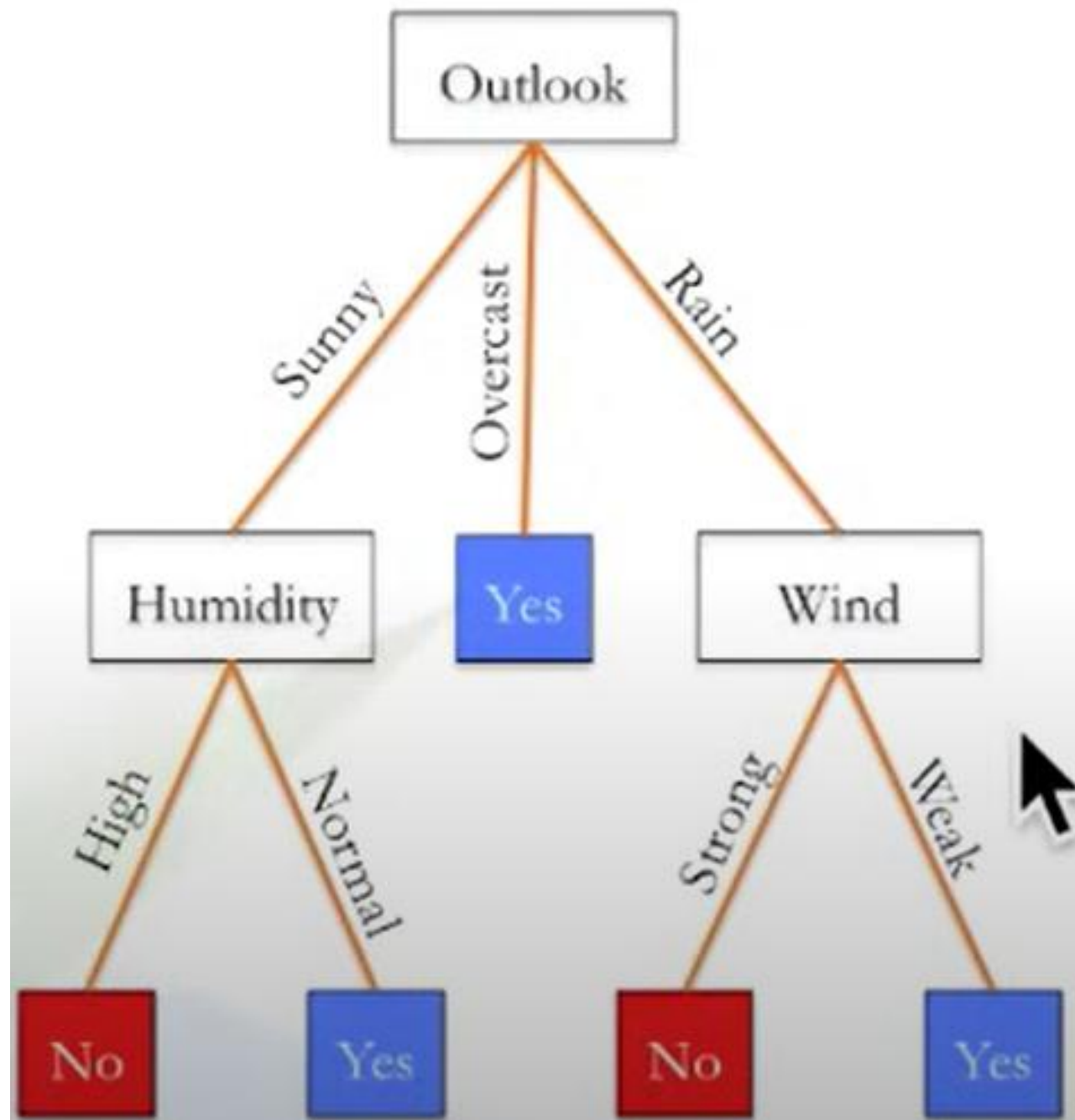
Information gain measures how well a feature splits the data by calculating the reduction in uncertainty (entropy) after the split. It helps the decision tree choose the attribute that best separates the classes.

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v)$$

- ❖ *S : the set of examples*
- ❖ *A : the attribute (feature) to split on*
- ❖ *S_v : the subset of S where attribute A has value v*

$$\text{Gain}(S, \text{Humidity}) = 0.940 - \left(\frac{7}{14} \cdot 0.985 + \frac{7}{14} \cdot 0.592 \right)$$

$$= 0.940 - (0.5 \cdot 0.985 + 0.5 \cdot 0.592) = 0.940 - (0.4925 + 0.296) = 0.940 - 0.7885 = 0.1515$$

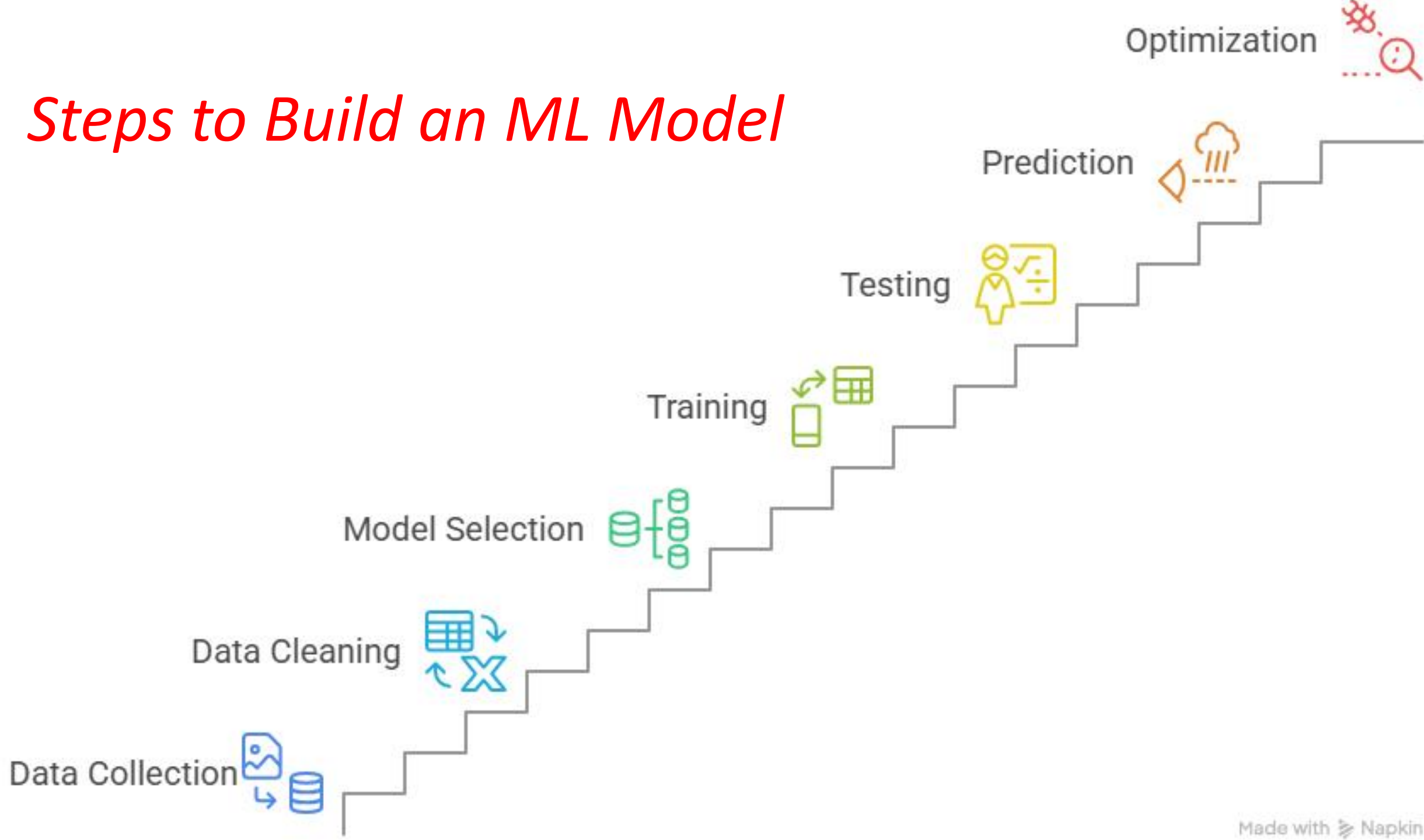


1. *Do you need to understand every internal process of the ML algorithms you use?*
2. *Do you have to write each ML algorithm by yourself?*
3. *What about the lab for developing ML-based IDS?*

Reminder

- Basics of AI, ML (supervised vs unsupervised)
- Decision Tree concept
- Steps to Build an ML Model

Steps to Build an ML Model



Implementation

Dataset preparation

1. Download the following datasets:



Google Colab Introduction

1. Cloud-based Python notebook
2. Free access to GPU/TPU
3. Great for beginners

Task 1: Create a Google Colab Account

Introduction to Advanced IA Techniques for IDS

In this module, we explore 4 essential AI techniques used to design smart IDS architectures:

1. Classification

Classification

Definition: Supervised learning technique used to predict a label (class) for each input.

Goal: Learn from labeled data to classify new observations as "Benign" or "Attack".

Examples: Decision Tree, k-Nearest Neighbors (k-NN), Random Forest, ...

Use Case in IDS: Detect whether a given packet or flow is malicious or not.

Introduction to Advanced IA Techniques for IDS

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1. Classification
2. Anomaly Detection

Anomaly Detection

Use Case in IDS: Detect zero-day attacks or rare attack types not seen during training.

Definition: Unsupervised or semi-supervised technique to detect data points that significantly deviate from the norm.

Goal: Identify unusual or suspicious patterns in data.

Examples: One-Class SVM, Isolation Forest, Autoencoders

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1. Classification
2. Anomaly Detection
3. Ensemble Learning

Ensemble Learning

Use Case in IDS: A context where there is a different modalities.

Definition: Combining predictions from multiple models to improve overall performance (**Voting** (Majority, Average), **Stacking**).

Goal: Boost accuracy

Types: Random Forest, XGBoost, ..

Introduction to Advanced IA Techniques for IDS

In this module, we explore 4 essential AI techniques used to design smart IDS architectures:

1. Classification
2. Anomaly Detection
3. Ensemble Learning
4. Federated Learning

Federated Learning

Use Case in IDS: Each charging station trains its own local model; only model updates are sent to the central server.

Definition: Collaborative machine learning approach where models are trained locally on devices and only model updates are shared.

Goal: Preserve data privacy and reduce communication cost.

Summary

These four techniques help build flexible and intelligent IDS architectures:

- Classification: Base method to label data
- Anomaly Detection: Identify unknown threats
- Ensemble Learning: Combine diverse models
- Federated Learning: Train models without sharing data